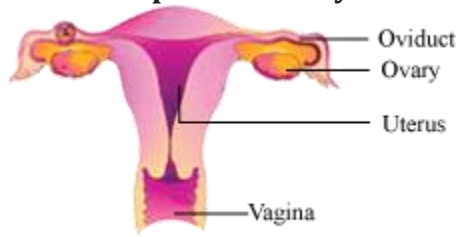


Reproduction in Animals

- **Male reproductive system:** It consists of testis, sperm duct and penis.

- Testes are involved in the production of male gametes called sperms.
- Millions of sperms are produced by the testes.
- Each sperm consists of three parts: Head, middle piece and tail.

- **Female reproductive system:**



Female reproductive system

- It consists of ovaries, oviduct and uterus.
 - Ovaries produce ova or eggs.
 - A single matured egg is released from ovary into oviduct every month.
 - Baby develops in the uterus.
 - Egg is also single celled like sperm.
- **Fertilization**
 - The process of fusion of male and female gametes (egg and sperm) to form zygote is known as **fertilization**.
 - It is of two types- Internal and external.
 - **Internal fertilization:** In this, the fusion of sperm and egg takes place inside the female's body. For example, cows, dogs and humans.
 - **External fertilization:** In this, the fusion of sperm and egg takes place outside the female's body in a surrounding medium, generally water. It occurs in frogs, fishes, starfish, etc.
- **Test tube baby**
 - A baby conceived by fertilization that occurs outside the mother's body is called test tube baby.
- **Development of the embryo**
 - The **zygote** repeatedly divides to form a ball of cells.

- The ball of cells then starts differentiating into tissues and organs. At this stage, it is called **embryo**.
- Embryo gets attached to the wall of the uterus and develops various body parts such as hands and legs.
- **Foetus** is a stage of embryo that shows main recognizable feature of mature organism.
- Foetus develops for nine months inside the mother's womb and is finally delivered.
- **Fertilization**- fusion of the nucleus of the sperm with the ovum to form a zygote. It occurs in the fallopian tube of females.
- Zygote divides to form an embryo.
- Embryo is implanted in the uterus.
- Foetus develops inside the mother's body for nine months (gestation period).

- **Oviparous animals**

- The animals that lay eggs are called **oviparous animals**. The examples include all kinds of birds, lizards, snakes, and frogs.

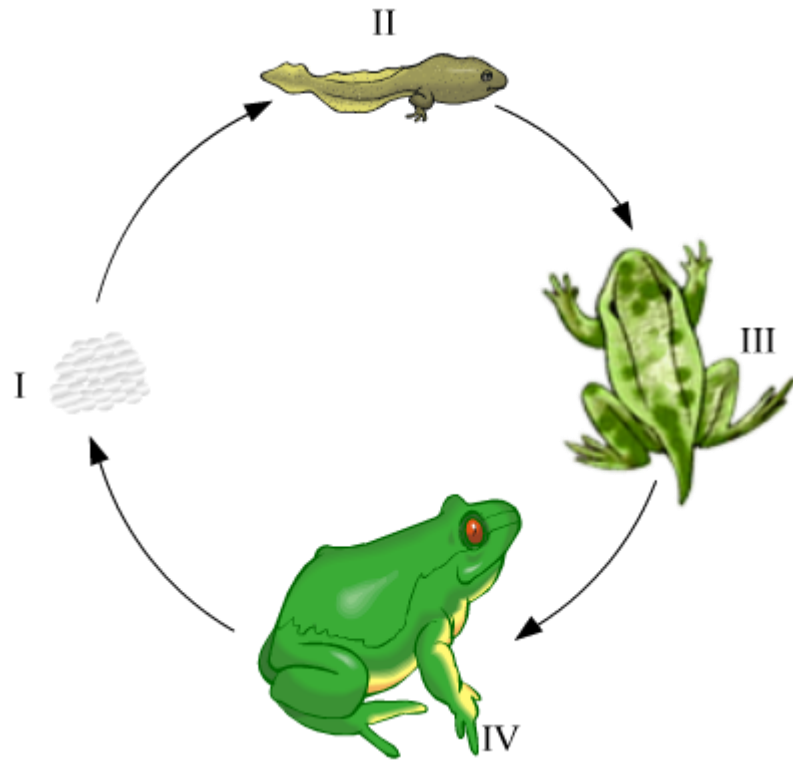
- **Viviparous animals**

- The animals that give birth to young ones are called **viviparous animals**. The examples include cows, dogs, and humans.

- **Metamorphosis**

- The biological process of transformation of larva into an adult is known as metamorphosis.
- The life cycle of **frog** consists of the following stages:

Egg → Tadpole (larva) → Adult

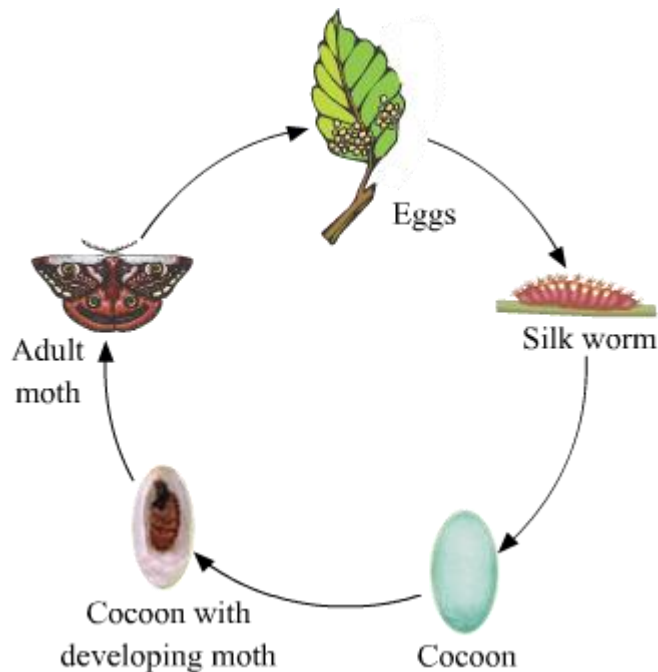


- **Hormones controlling metamorphosis in frogs**

- Thyroxin (produced by the thyroid gland) initiates the process of a tadpole's metamorphosis into an adult frog.
- In the absence of thyroxin, the tadpole does not transform into an adult and remains in the tadpole stage.

Life cycle of silkworm

- Silkworm grows on mulberry trees and feeds on its leaves. During a stage in its life cycle, silkworm spins a cocoon around itself. Silk is obtained from this cocoon. The life cycle of a silkworm is represented below,



- **Hormone responsible for metamorphosis in insects**

- In insects, metamorphosis is controlled by the insect hormones. Some of the insect hormones are:
 1. Prothoracicotrophic Hormone (PTTH)
 2. Ecdysone
 3. Juvenile Hormone (JH)

- **Asexual reproduction**

- The type of reproduction which involves only a single parent and the new individuals are formed without the fusion of gametes is known as **asexual reproduction**.
- Two common methods of asexual reproduction are:
 - **Budding** - It involves the formation of new individual from the bulging of the parent body. This phenomenon is very common in plants, fungi and animals such as *Hydra* and yeast.
 - **Fission- Binary fission** is the type of asexual reproduction that occurs in *Amoeba*. It is a type of asexual reproduction in which a single cell divides into two halves.

- **Cloning**

- Cloning is the process used to create an exact copy of a cell, tissue or an organism. Dolly, a sheep was the first mammal to be cloned. Dolly was cloned by Ian Wilmut and his colleagues in 1996.